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Department of Artificial Intelligence and Data Science

Project Work Phase – I

First Review

HIERARCHICAL EXPLAINABLE ENSEMBLE CLINICAL RISK PREDICTION SYSTEM USING MULTIMODAL DEEP LEARNING

Batch No.: 18

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Problem Statement(s)

ICUs produce extensive patient data, such as demographics, vital signs, lab results, and clinical records. Clinicians need to swiftly analyze this data to detect worsening conditions, as delays can have severe consequences.

- **Single-Modality Learning**

Models often rely on a single data type, missing the comprehensive view clinicians use, resulting in less accurate.

- **Independent Feature Processing**

Models may not recognize interactions between data features, overlooking critical risk indicators.

- **Lack of Explainability**

Deep learning models often act as "black boxes," providing predictions without clear explanations, which reduces trust.

- **Fixed Ensemble Methods**

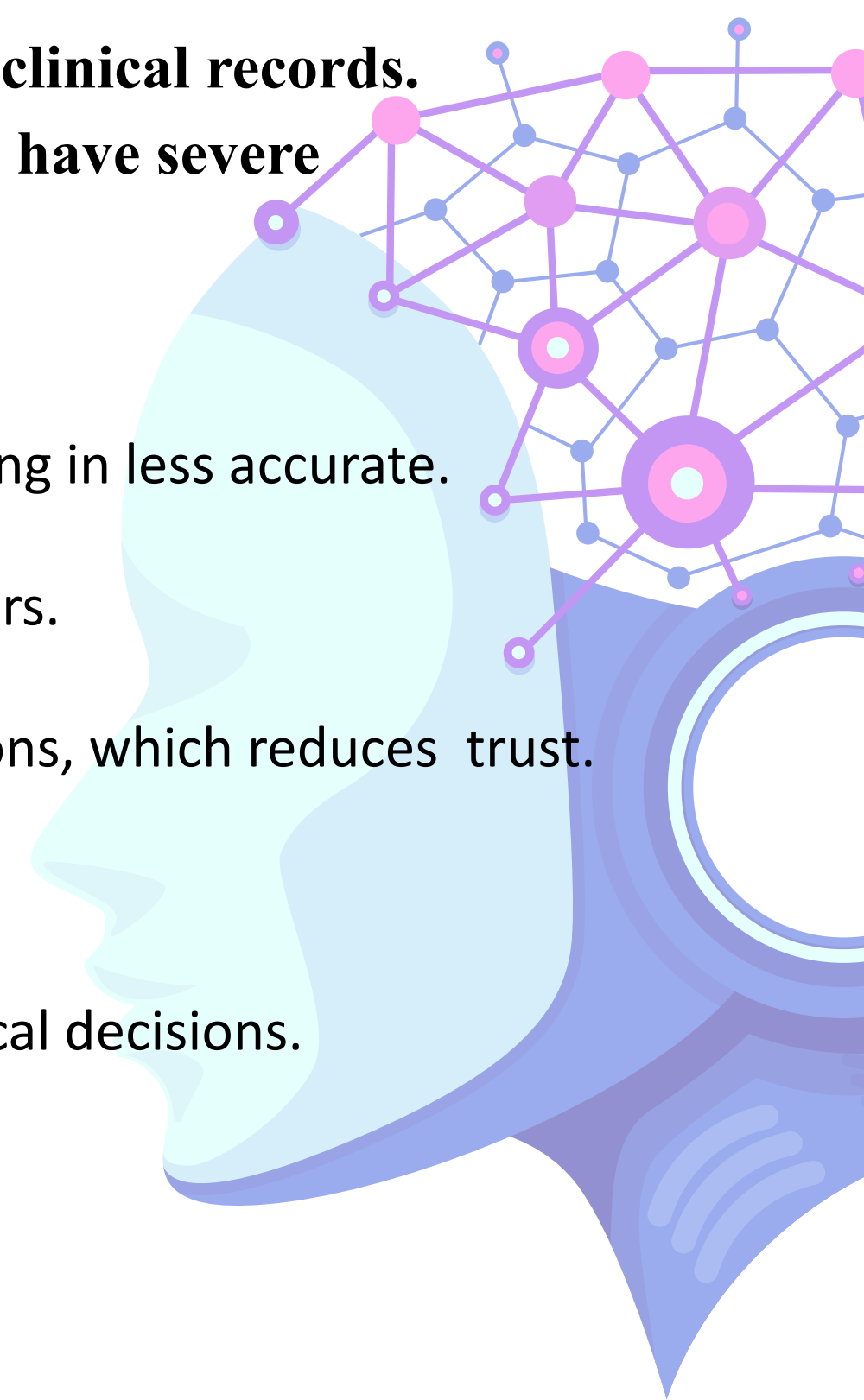
Traditional models use fixed weights and cannot adapt to individual patient needs.

- **No Prediction Confidence**

AI systems usually offer probability scores without indicating reliability, which is crucial for clinical decisions.

- **Poor Generalization**

Models tested on a single dataset may not perform well across different healthcare settings.



Objective(s)

01 - Multimodal Clinical Feature Learning(A)

Extract meaningful representations from demographic data, laboratory measurements, and time-series vital signs using specialised deep learning models.

02 - Autonomous AI Agent Development

Transform TabNet, Bi-LSTM, and CNN models into independent AI agents capable of analysing patient data from distinct clinical perspectives.

03 - Multi-Agent Collaborative Consensus

Enable agents to communicate, debate, and justify their predictions to achieve a reliable clinical risk assessment through collaborative reasoning.

04 - Explainable Decision Intelligence

Provide transparent explanations using SHAP analysis, agent reasoning traces, attention visualizations, and counterfactual recommendations.

05 - Uncertainty-Aware Risk Prediction

Incorporate confidence estimation and uncertainty quantification to identify ambiguous cases requiring clinician review.

06 - Clinical Validation and Deployment

Evaluate the framework on MIMIC-III and eICU datasets and deploy an interactive dashboard for real-time clinical decision support.

SDG Alignment

SDG	Aim	Impact
3 – Good Health and Well-Being	Better health	Early prediction and improved decisions
9 – Industry, Innovation and Infrastructure	Stronger systems	Smarter tools with AI and explainability
10 – Reduced Inequalities	Fair access	Scalable, open tech for inclusive analytics

Literature Review



Paper	Author & Year	Method	Outcome	Limitation
Timely ICU Outcome Prediction Utilizing Stochastic Signal Analysis and Machine Learning Techniques with Readily Available Vital Sign Data <i>(IEEE JBHI)</i>	Li et al. (2024)	Stochastic Signal Analysis + Machine Learning using Vital Signs	Early ICU outcome prediction with improved accuracy using continuously monitored vital signs.	Uses mainly vital signs ; does not integrate laboratory and demographic data. Limited explainability and no uncertainty estimation.
TransformerFusionNet: A Real-Time Multimodal Framework for ICU Heart Failure Mortality Prediction Using Big Data Streaming <i>(IEEE ICCA)</i>	Zhang et al. (2024)	Transformer-based Multimodal Fusion	Improved real-time mortality prediction for ICU heart failure patients using streaming clinical data.	Disease-specific (heart failure), lacks adaptive ensemble weighting and comprehensive explainability.
Predicting ICU Admissions using Interpretable Machine Learning <i>(IEEE Conference)</i>	Kumar et al. (2024)	Interpretable Machine Learning	Improved prediction of ICU admission while providing feature-level explanations.	Predicts ICU admission rather than mortality; limited multimodal integration and deep feature learning.
Scalable and Accurate Deep Learning with Electronic Health Records	Rajkomar et al. (2018)	Deep Learning on Multimodal EHR	Demonstrated high-performance clinical prediction from EHR data.	Black-box predictions with limited explainability and confidence estimation.
TabNet: Attentive Interpretable Tabular Learning	Arik & Pfister (2021)	TabNet	Achieved interpretable learning for structured clinical data.	Primarily designed for tabular data; does not effectively model temporal vital-sign information.
A Unified Approach to Interpreting Model Predictions (SHAP)	Lundberg & Lee (2017)	SHAP Explainable AI	Provided global and local explanations for AI predictions.	Improves interpretability only; does not enhance prediction performance or uncertainty estimation.

Research Gap

- Problem 1 — Huge Volume of ICU Data
- Problem 2 — Single-Modality Models
- Problem 3 — Black-Box Models
- Problem 4 — Poor Generalization
- Problem 5 — No Confidence Estimation

Gap Addressed by Our Work

We propose a **Hierarchical Explainable Ensemble Clinical Risk Prediction System** that combines multimodal deep learning, collaborative ensemble learning, hierarchical explainability (**SHAP, attention, counterfactuals**), and uncertainty estimation to provide accurate, transparent, and reliable clinical risk predictions.

How Your Project Addresses These Gaps

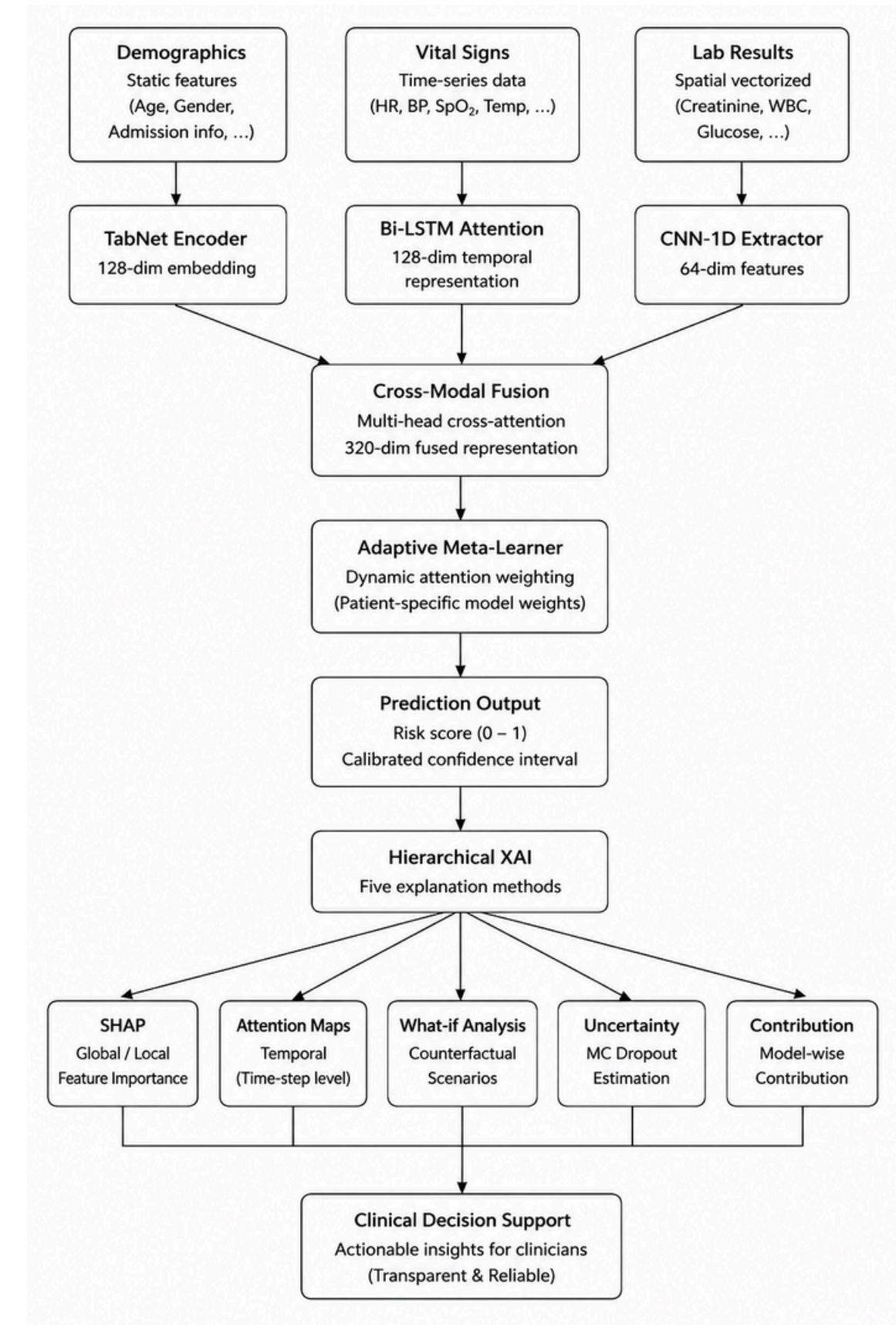
Existing Research Gap	EECRPS-XAI Solution
Single data modality	Multimodal learning (demographics + vitals + labs)
Separate feature processing	Cross-Modal Attention Fusion
Fixed ensemble methods	Adaptive patient-specific meta-learner
Black-box predictions	Global SHAP, Local SHAP, Temporal Attention, Counterfactuals
No uncertainty estimation	Monte Carlo Dropout confidence intervals
Single-dataset evaluation	MIMIC-III, eICU, PhysioNet 2019, and Synthea

Existing Technology

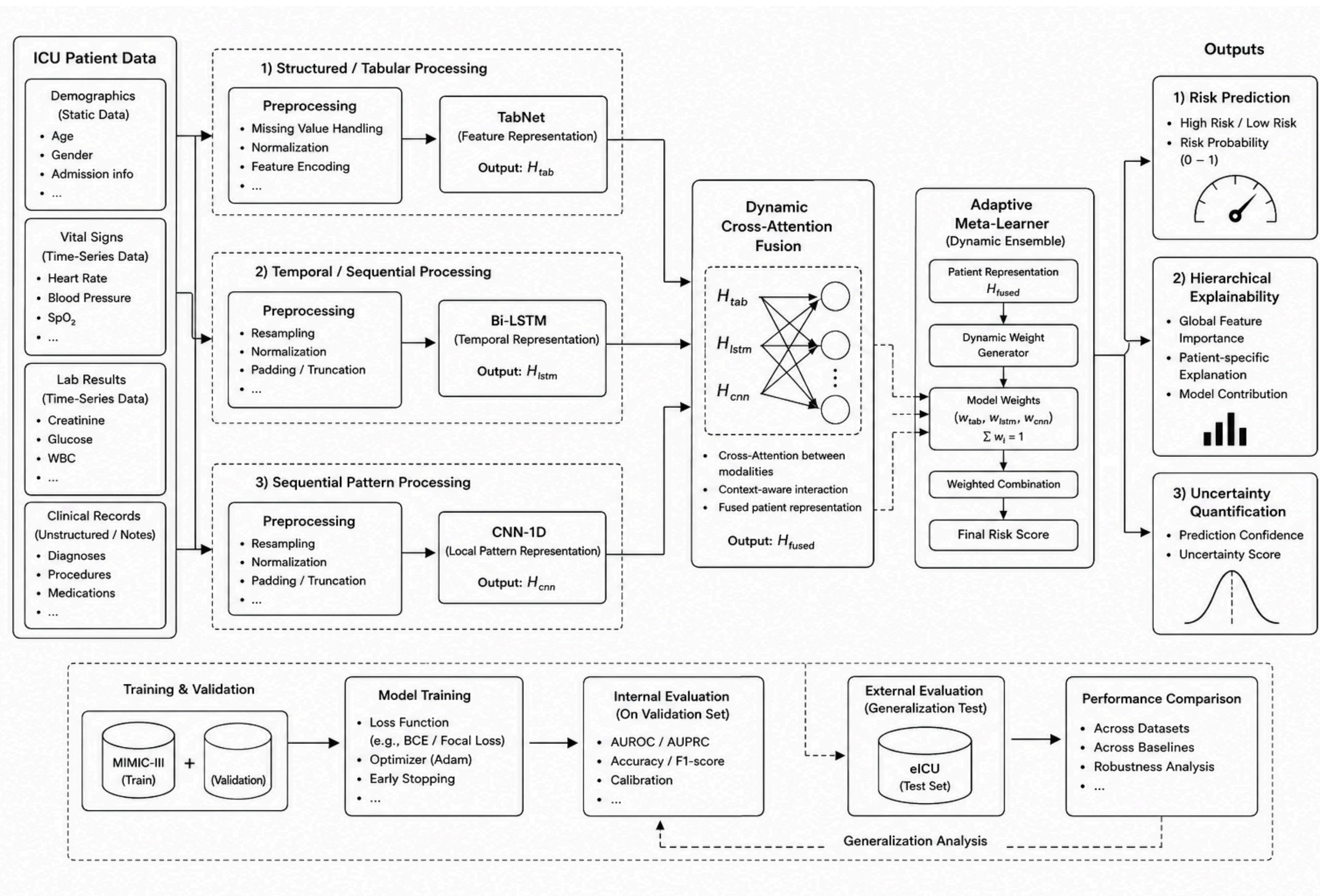
- **Current Solutions & Limitations:** Current clinical methods isolate medical data types using standalone models or basic averaging ensembles, creating opaque "black boxes" that lack vital explanations and prediction confidence intervals.
- **Technological Trends:** Healthcare AI is shifting toward multi-agent systems where specialized AI agents collaborate to analyze multimodal patient data, explain their reasoning, and quantify uncertainty, enabling more trustworthy and intelligent clinical decision support.

Proposed Work/ Methodology

- **Concept and Approach:** EECRPS integrates TabNet, Bi-LSTM, and CNN-1D models using dynamic cross-attention fusion and an adaptive meta-learner for patient risk prediction.
- **Innovations:** The framework incorporates dynamic ensemble weighting, hierarchical explainability, and uncertainty quantification for transparent and reliable predictions.
- **Feasibility:** The project leverages open-source datasets (MIMIC-III, eICU) and Python-based AI frameworks, requiring no additional data acquisition costs.



Architecture Diagram



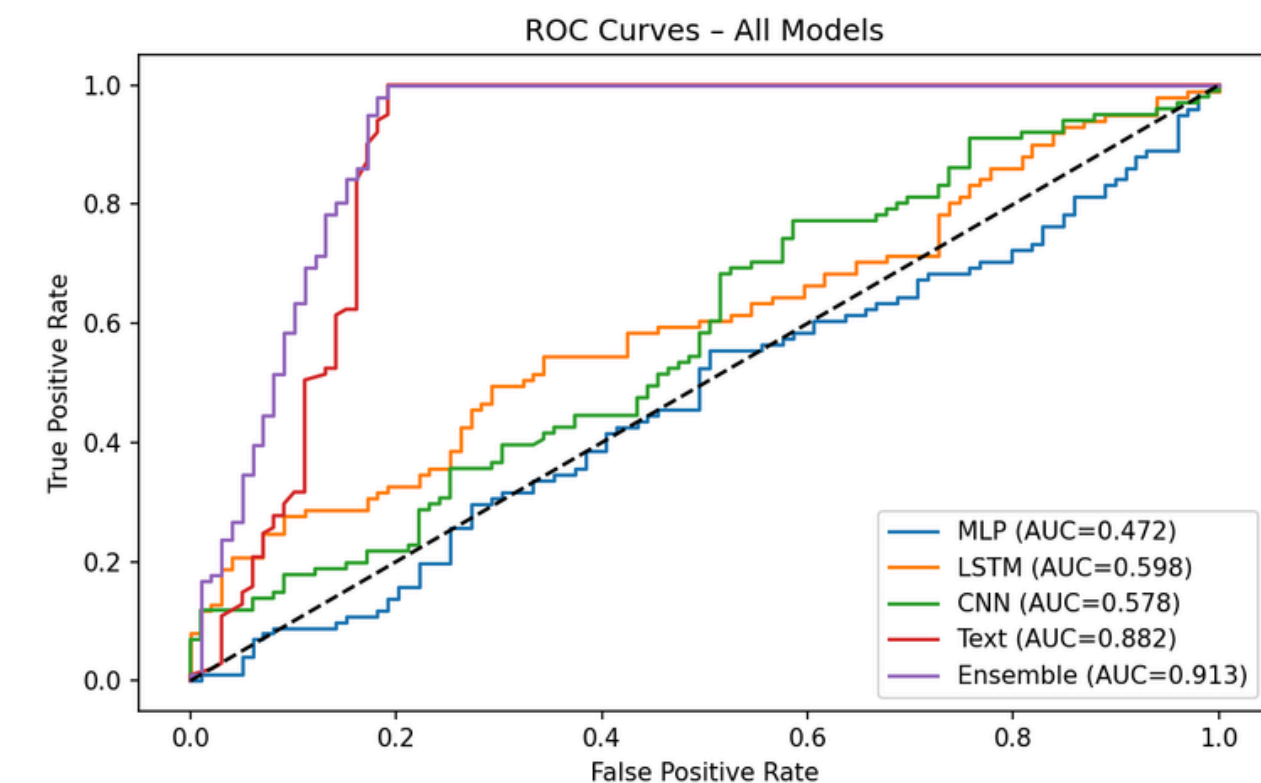
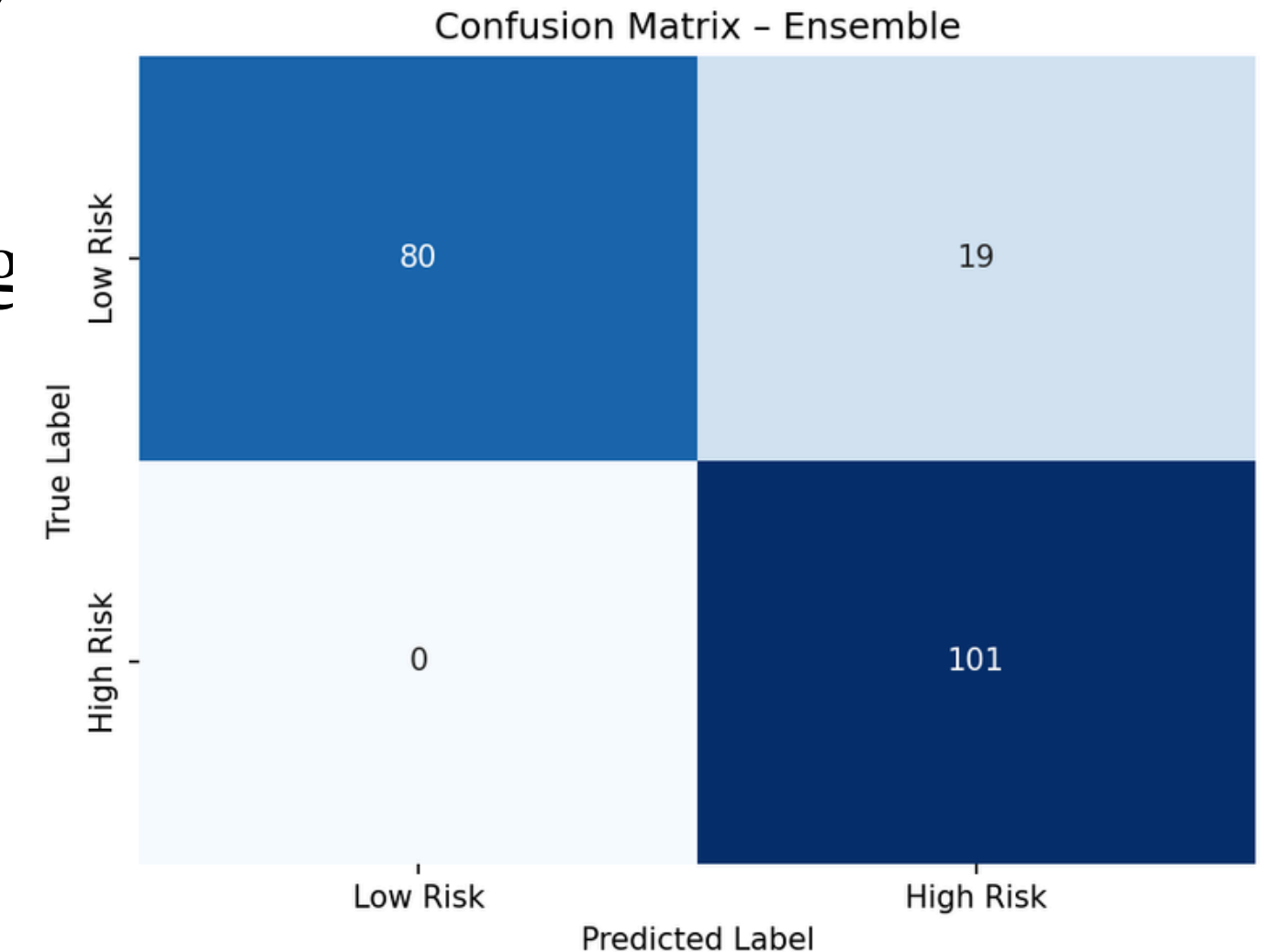
Legal Open-Source Datasets Used for Clinical Risk Prediction



Dataset	Type	Size	License	Access Link
MIMIC-III	EHR Time-Series + Tabular Data	40,000+ ICU Patients (Vitals, Labs, Demographics)	PhysioNet Credentialed Health Data	physionet.org/content/mimiciii
eICU Collaborative	Multi-Center ICU Data	200,000+ Admissions from 335 Hospitals	PhysioNet Credentialed	physionet.org/content/eicu-crd
PhysioNet Challenge 2019	Sepsis Prediction Benchmark	40,336 Patients with Hourly Vitals	Open Data Commons	physionet.org/content/challenge-2019
Synthea Synthetic EHR	Synthetic Patient Records	Unlimited Synthetic Data	Apache 2.0 (Open Source)	synthea.mitre.org

Expected Outcome~

- **Expected Results:** Achieve >0.95 AUROC and >0.70 AUPRC in mortality prediction on MIMIC-III, producing 4-level clinical explanations (SHAP, temporal attention, counterfactuals) alongside calibrated safety confidence intervals.
- **Potential Benefits:** Reduces diagnostic delays by identifying hidden multi-modal risk patterns, mitigates single-model biases, and builds clinician trust through transparent, actionable "what-if" insight generation.
- **Clinical Applications:** Powers real-time ICU decision support dashboards for early sepsis or mortality triage, scales cross-hospital monitoring via eICU validation, and acts as an interpretable baseline for longitudinal medical research.



Reference(s)

No.	Authors	Title	Venue	Year	Category
1	Johnson et al.	MIMIC-III: A freely accessible critical care database	Scientific Data	2016	Dataset
2	Rajkomar et al.	Scalable and accurate deep learning with EHR	NPJ Digital Medicine	2018	Deep Learning
3	Arik & Pfister	TabNet: Attentive Interpretable Tabular Learning	AAAI	2021	Architecture
4	Lundberg & Lee	Unified approach to interpreting model predictions	NeurIPS	2017	XAI
5	Mothilal et al.	Diverse Counterfactual Explanations	ACM FAccT	2020	XAI
6	Gal & Ghahramani	Dropout as Bayesian Approximation	ICML	2016	Uncertainty
7	Zhang & Ma	Ensemble Deep Learning for Health Informatics	IEEE Rev. Biomed. Eng.	2021	Ensemble
8	Tjoa & Guan	Survey on Explainable AI in Healthcare	IEEE Access	2021	XAI
9	Chawla et al.	SMOTE: Synthetic Minority Over-sampling	JAIR	2002	Imbalanced Learning